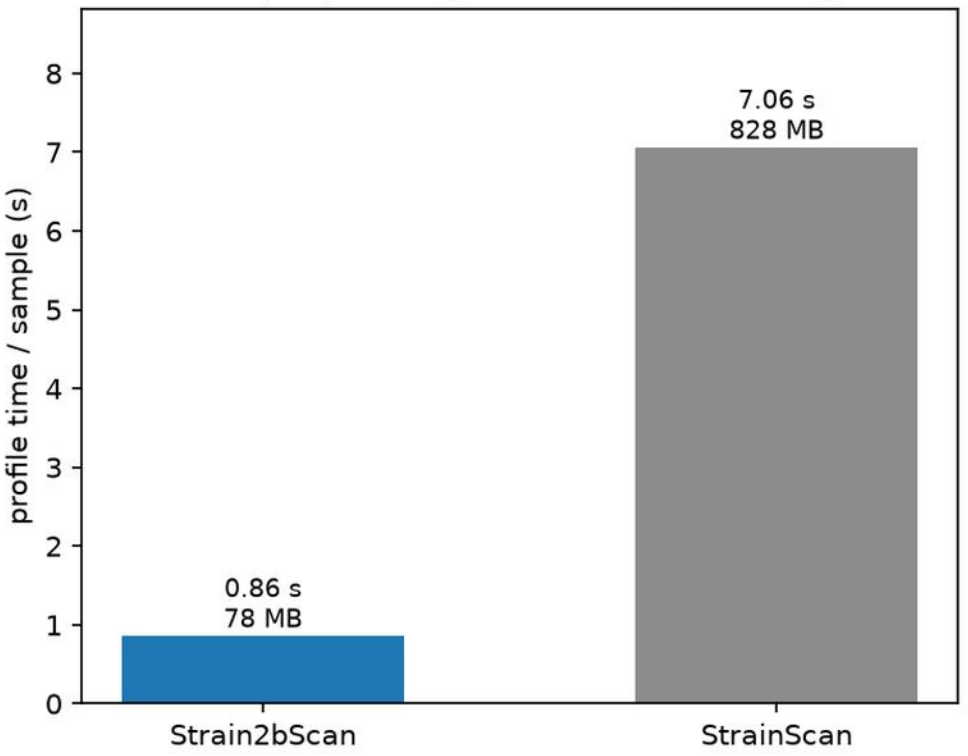
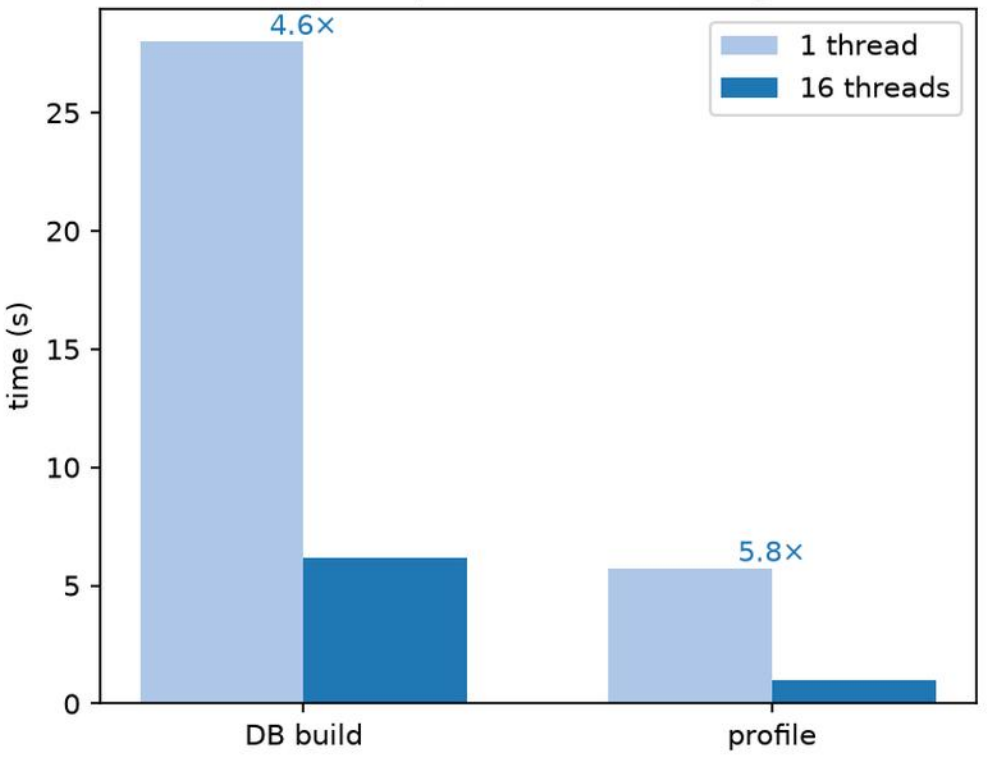
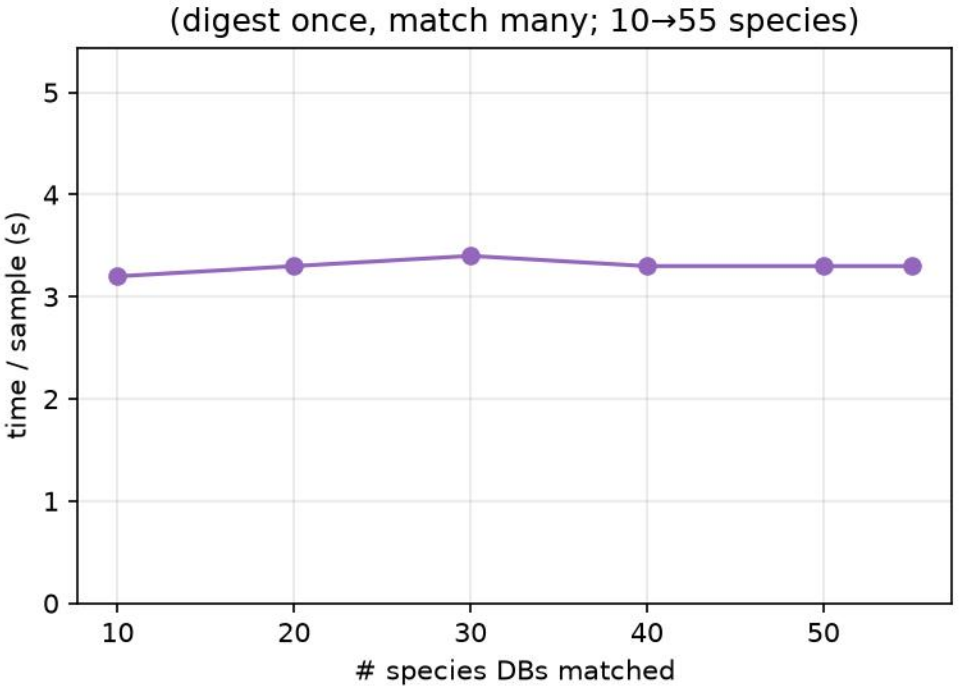
Per-sample profiling: ~8× faster, ~11× lighter



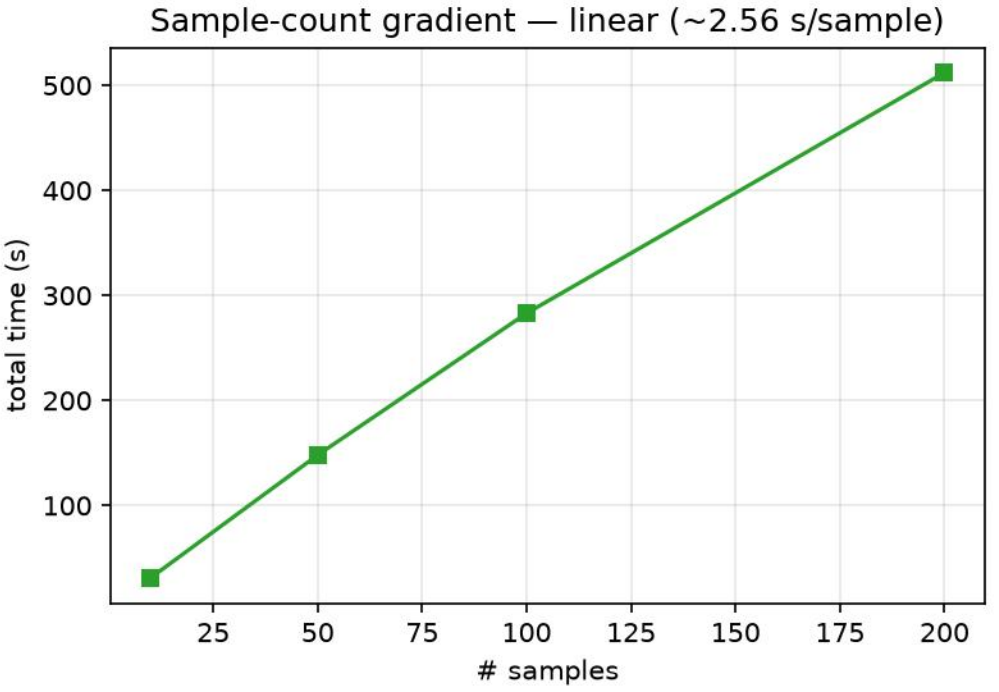
Parallel speedup (real C. acnes, 64 genomes)



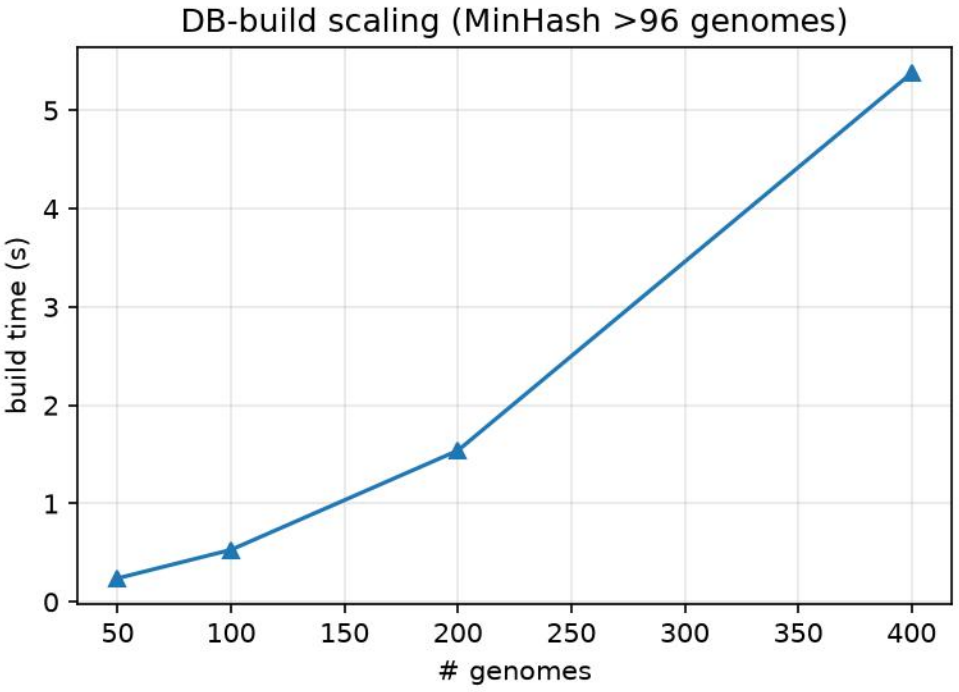
Species-count gradient — flat
(digest once, match many; 10→55 species)



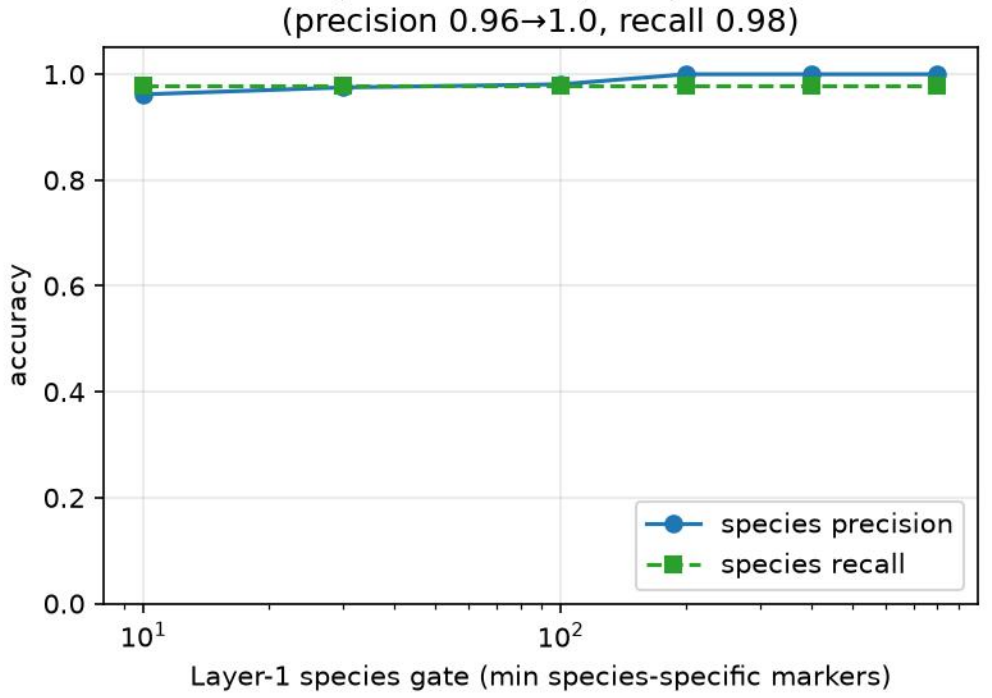
Sample-count gradient — linear (~2.56 s/sample)



DB-build scaling (MinHash >96 genomes)



Multi-species accuracy vs species gate
(precision 0.96→1.0, recall 0.98)



Community-scale strain profiling (55 real species)

